

Chapter 8: Magnetism

Key Concepts

1. A magnet is an object that produces a magnetic field and attracts ferromagnetic materials like iron, nickel, and cobalt.
2. Every magnet has two poles: North Pole and South Pole. Like poles repel, unlike poles attract.
3. Magnetic field lines are imaginary lines that represent the direction and strength of a magnetic field. They run from North to South outside the magnet.
4. Magnetic materials can be classified as ferromagnetic (strongly attracted), paramagnetic (weakly attracted), and diamagnetic (weakly repelled).
5. Induced magnetism occurs when a magnetic material becomes magnetized when placed near a magnet.
6. Electromagnetism is the production of a magnetic field by an electric current flowing through a conductor.
7. A current-carrying conductor produces a circular magnetic field around it. The direction is given by the Right-Hand Rule.
8. An electromagnet is a coil of wire wound around a soft iron core that becomes magnetic only when current flows through it.
9. The strength of an electromagnet depends on the number of turns, current magnitude, and the type of core material.
10. An electric motor converts electrical energy into mechanical energy using the interaction between magnetic fields and current-carrying conductors.
11. An electric generator converts mechanical energy into electrical energy using electromagnetic induction.
12. A magnetic compass uses a freely rotating magnetized needle to align with the Earth's magnetic field.
13. The Earth behaves like a giant bar magnet with its magnetic South Pole near the geographic North Pole.
14. Magnets are used in speakers, hard drives, MRI machines, electric motors, generators, and magnetic separators.
15. The magnetic field is strongest at the poles and weakest at the center of a magnet.

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MCQ's

1. Which of the following is a natural magnet?

A. Alnico B. Ferrite C. Lodestone D. Steel

Answer: C

2. The strength of a magnetic field is maximum at:

A. Center of magnet B. North Pole only C. South Pole only D. Both poles

Answer: D

3. Like magnetic poles always:

A. Attract each other B. Repel each other C. Cancel each other D. Induce each other

Answer: B

4. The magnetic field lines outside a bar magnet run from:

A. South to North B. North to South C. East to West D. West to East

Answer: B

5. Which material is best suited for making a temporary electromagnet?

A. Steel B. Soft iron C. Cobalt D. Nickel

Answer: B

6. The direction of the magnetic field around a straight current-carrying conductor is given by:

A. Fleming's Left-Hand Rule B. Right-Hand Thumb Rule C. Fleming's Right-Hand Rule D. Lenz's Law

Answer: B

7. An electric motor works on the principle of:

A. Electromagnetic induction B. Magnetic effect of current C. Electrostatic force D. Thermal effect of current

Answer: B

8. Which device converts mechanical energy into electrical energy?

A. Electric motor B. Electric generator C. Transformer D. Electromagnet

Q7. What is the principle of an electric motor?

Ans: An electric motor works on the principle that a current-carrying conductor placed in a magnetic field experiences a force (mechanical force). This force causes the coil to rotate, converting electrical energy into mechanical energy.

Q8. Distinguish between an electric motor and an electric generator.

Ans: An electric motor converts electrical energy into mechanical energy and works on the magnetic effect of current. An electric generator converts mechanical energy into electrical energy and works on the principle of electromagnetic induction.

Q9. Why does a magnetic compass needle point north-south?

Ans: The Earth itself acts like a giant bar magnet with its magnetic South Pole near the geographic North Pole. The compass needle (a small magnet) aligns itself with the Earth's magnetic field, so its North Pole points towards the geographic North Pole.

Q10. Write any three practical uses of magnets in daily life.

Ans: (i) In loudspeakers and headphones to convert electrical signals into sound. (ii) In hard disk drives to store data magnetically. (iii) In MRI machines to produce detailed images of internal body organs.

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Long Questions

Q1. Describe in detail the properties of magnets and the characteristics of magnetic field lines.

Ans: Magnets have the following properties:

- **Attractive property:** Magnets attract ferromagnetic materials such as iron, nickel, and cobalt.
- **Directive property:** A freely suspended magnet always aligns itself in the north-south direction.
- **Polar property:** Every magnet has two poles — North and South. Like poles repel, unlike poles attract.
- **Inductive property:** A magnet can induce magnetism in nearby magnetic materials.
- **Pair property:** Poles always exist in pairs. If a magnet is broken, each piece becomes a complete magnet with both poles.

Magnetic field lines have these characteristics:

- They emerge from the North Pole and enter the South Pole outside the magnet. Inside the magnet, they run from South to North, forming closed loops.
- The tangent at any point on a field line gives the direction of the magnetic field at that point.
- They never intersect each other. If they did, it would mean two directions of the field at the same point, which is impossible.
- The density of field lines indicates the strength of the magnetic field — closer lines mean a stronger field.
- The field is strongest near the poles where the lines are most concentrated.

Q2. Explain the phenomenon of electromagnetism. Describe the magnetic field pattern around a straight conductor and a solenoid.

Ans: Electromagnetism is the production of a magnetic field by an electric current. In 1820, Hans Christian Oersted discovered that a current-carrying wire deflects a nearby compass needle, proving that electricity and magnetism are related.

Magnetic field around a straight conductor: When current flows through a straight conductor, concentric circular magnetic field lines are produced around it. Using the Right-Hand Thumb Rule, if the thumb points in the direction of current, the curled fingers show the direction of the magnetic field. The field strength decreases as the distance from the conductor increases.

Magnetic field of a solenoid: A solenoid is a long coil of wire with many turns. When current passes through it, the magnetic field inside the solenoid is strong and uniform (parallel and equally spaced). The field outside the solenoid is weak and similar to that of a bar magnet. One end of the solenoid acts as North Pole and the other as South Pole.

Q3. Describe the construction, working, and applications of an electromagnet.

Ans: An electromagnet is a temporary magnet made by winding a coil of insulated copper wire around a soft iron core.

Construction: A soft iron rod (core) is taken. An insulated copper wire is wound tightly around it in the form of a coil. The two ends of the wire are connected to a battery through a switch.

Working: When current flows through the coil, the soft iron core becomes magnetized and attracts

Thus, the magnetic field strength is 1.0×10^{-5} Tesla.

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Q6. A wire of length 0.5 m moves with a velocity of 10 m/s perpendicular to a magnetic field of 0.2 T. Calculate the induced emf.

Solution:

The induced emf in a moving conductor is:

$$\epsilon = B L v$$

$$\epsilon = 0.2 \times 0.5 \times 10$$

$$\epsilon = 1.0 \text{ V}$$

Thus, the induced emf is 1.0 V.

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Q7. The magnetic flux through a coil of 100 turns changes from 0.05 Wb to 0.01 Wb in 0.2 s. Calculate the average induced emf.

Solution:

Using Faraday's law:

$$\epsilon = -N \left(\frac{\Delta \Phi}{\Delta t} \right)$$

$$\Delta \Phi = \Phi_{\text{final}} - \Phi_{\text{initial}} = 0.01 - 0.05 = -0.04 \text{ Wb}$$

$$\Delta t = 0.2 \text{ s}$$

$$\epsilon = -100 \times \left(\frac{-0.04}{0.2} \right)$$

$$\epsilon = 100 \times 0.2$$

$$\epsilon = 20 \text{ V}$$

Thus, the average induced emf is 20 V.

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Q8. An electromagnet has a solenoid with 800 turns and length 0.4 m. If it produces a magnetic field of 0.025 T, calculate the current flowing through it. ($\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$)

Solution:

$$B = \mu_0 n I$$

$$n = N/L = 800 / 0.4 = 2000 \text{ turns/m}$$

$$0.025 = (4\pi \times 10^{-7}) \times 2000 \times I$$

$$0.025 = 8000\pi \times 10^{-7} \times I$$

$$0.025 = 8\pi \times 10^{-4} \times I$$

$$I = 0.025 / (8\pi \times 10^{-4})$$

$$I = 0.025 / (2.513 \times 10^{-3})$$

$$I = 9.95 \text{ A}$$

Thus, the current required is approximately 9.95 A.